

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 9

Sequences & Series

Duration: 2.5 hours Total Marks: 100
Name: _____ Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Find the 5th term of sequence: 2, 5, 8, 11, ... [2 marks]

2. Find common ratio of GP: 3, 6, 12, 24, ... [2 marks]

3. Sum of arithmetic series: $1 + 2 + 3 + \dots + 100$ [3 marks]

4. If first term $a = 2$ and common difference $d = 3$, find 10th term [2 marks]

5. Sum of GP with first term 1, ratio 2, 5 terms [3 marks]

6. Is 0.333... a GP or AP? [2 marks]

7. Find sum to infinity of GP: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ [2 marks]

8. Find sum of: $5 + 10 + 15 + \dots + 100$ [3 marks]

SECTION B: Long Answer Questions (60 marks)

9. A sequence has $a_1 = 3$, $a_{n+1} = 2a_n - 1$. Find a_2 , a_3 , a_4 . (4 marks)

10. Sum of first n terms of AP is $S_n = n^2 + 2n$. Find first 3 terms. (5 marks)

11. Geometric series: sum to n terms is $S_n = 3(2^n - 1)$. Find a_1 and r . (4 marks)

12. How many terms of AP (2, 5, 8, ...) sum to 155? (5 marks)

13. Prove: Sum of AP = $\frac{n}{2}(2a + (n-1)d)$. (4 marks)

14. A bouncing ball reaches 80% of previous height. If initial height is 10m, find total distance traveled. (5 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: 5th term of 2,5,8,11,...

$$d = 3, a_5 = a_1 + 4d = 2 + 12 = 14$$

Q2: Common ratio of 3,6,12,24,...

$$r = 6/3 = 2$$

Q3: Sum 1+2+...+100

$$S_n = n/2(\text{first} + \text{last}) = 100/2(1 + 100) = 50 \times 101 = 5050$$

Q4: 10th term with a=2, d=3

$$a_{10} = a + 9d = 2 + 27 = 29$$

Q5: Sum of GP: a=1, r=2, n=5

$$S_n = a(r^n - 1)/(r - 1) = 1(32 - 1)/1 = 31$$

Q6: Is 0.333... GP or AP?

$$0.333... = 1/3 + 1/30 + 1/300 + ... \text{ (infinite GP with } r=0.1)$$

Q7: Sum to infinity

$$S_\infty = a/(1-r) = 1/(1-0.5) = 2$$

Q8: Sum 5+10+...+100

$$a=5, d=5, n=20, S_n = n/2(2a + (n-1)d) = 10(10 + 95) = 1050$$

SECTION B SOLUTIONS

Q9: Recursive sequence

$$a_1=3, a_n=2(a_{n-1})-1=5, a_3=2(a_2)-1=9, a_4=2(a_3)-1=17$$

Q10: First 3 terms

$$S_1 = 1 + 2 = 3, \text{ so } a_1 = 3; S_2 = 4 + 4 = 8, \text{ so } a_2 = 5; S_3 = 9 + 6 = 15, \text{ so } a_3 = 7$$

Q11: Find a and r

$$S_2 = 3(2a - 1), S_3 = 3(2a - 1) = 3, \text{ so } a=3; S_4 = 3(4a - 1) = 9, a_4 = 9 - 3 = 6, r = 2$$

Q12: How many terms sum to 155

$$S_n = n/2(2a + (n-1)d) = n/2(4 + 3(n-1)) = 155, n(3n+1) = 310, 3n^2+n-310=0, n=10$$

Q13: Prove sum formula

AP: $a, a+d, a+2d, \dots, a+(n-1)d$, $S_n = a + (a+d) + \dots + (a+(n-1)d) = na + d(0+1+2+\dots+(n-1))$
 $= na + d \times n(n-1)/2 = n/2(2a+(n-1)d)$

Q14: Total distance

Heights: $10 + 8 + 6.4 + 5.12 + \dots = 10 + 2(8 + 6.4 + 5.12 + \dots) = 10 + 2 \times 8/(1-0.8) = 10 + 80$
 $= 90\text{m}$